

Meeting notes and Attendance

Attendance:

[illegible]

Meeting Notes:

Meeting 1 (13/1/26):

Reducing Carbon Footprint through interactive games

Users

- Students / children

Project Lead - Purni

Technical lead - Purni

Data lead - Purni

UI/UX lead - Purni

QA and testing - Purni

Documentation - AKK

Software - Bradley

Software - Ashley

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Advertise to parents about how it can help build child's carbon awareness

Can be used as a reward system to encourage children to save energy and reduce their carbon footprint

Helping reduce global carbon emissions

Must	Should	Could	Will Not
- Input environment Carbon Emission Calculator/Database - Progress Tracker (Dashboard) - Groups	- Competitions - Reward System	- 5.0V - 1.0V - 2.0V - 3.0V - 4.0V - 6.0V - 7.0V - 8.0V - 9.0V - 10.0V - 11.0V - 12.0V - 13.0V - 14.0V - 15.0V - 16.0V - 17.0V - 18.0V - 19.0V - 20.0V - 21.0V - 22.0V - 23.0V - 24.0V - 25.0V - 26.0V - 27.0V - 28.0V - 29.0V - 30.0V - 31.0V - 32.0V - 33.0V - 34.0V - 35.0V - 36.0V - 37.0V - 38.0V - 39.0V - 40.0V - 41.0V - 42.0V - 43.0V - 44.0V - 45.0V - 46.0V - 47.0V - 48.0V - 49.0V - 50.0V - 51.0V - 52.0V - 53.0V - 54.0V - 55.0V - 56.0V - 57.0V - 58.0V - 59.0V - 60.0V - 61.0V - 62.0V - 63.0V - 64.0V - 65.0V - 66.0V - 67.0V - 68.0V - 69.0V - 70.0V - 71.0V - 72.0V - 73.0V - 74.0V - 75.0V - 76.0V - 77.0V - 78.0V - 79.0V - 80.0V - 81.0V - 82.0V - 83.0V - 84.0V - 85.0V - 86.0V - 87.0V - 88.0V - 89.0V - 90.0V - 91.0V - 92.0V - 93.0V - 94.0V - 95.0V - 96.0V - 97.0V - 98.0V - 99.0V - 100.0V	- 1.0V - 2.0V - 3.0V - 4.0V - 5.0V - 6.0V - 7.0V - 8.0V - 9.0V - 10.0V - 11.0V - 12.0V - 13.0V - 14.0V - 15.0V - 16.0V - 17.0V - 18.0V - 19.0V - 20.0V - 21.0V - 22.0V - 23.0V - 24.0V - 25.0V - 26.0V - 27.0V - 28.0V - 29.0V - 30.0V - 31.0V - 32.0V - 33.0V - 34.0V - 35.0V - 36.0V - 37.0V - 38.0V - 39.0V - 40.0V - 41.0V - 42.0V - 43.0V - 44.0V - 45.0V - 46.0V - 47.0V - 48.0V - 49.0V - 50.0V - 51.0V - 52.0V - 53.0V - 54.0V - 55.0V - 56.0V - 57.0V - 58.0V - 59.0V - 60.0V - 61.0V - 62.0V - 63.0V - 64.0V - 65.0V - 66.0V - 67.0V - 68.0V - 69.0V - 70.0V - 71.0V - 72.0V - 73.0V - 74.0V - 75.0V - 76.0V - 77.0V - 78.0V - 79.0V - 80.0V - 81.0V - 82.0V - 83.0V - 84.0V - 85.0V - 86.0V - 87.0V - 88.0V - 89.0V - 90.0V - 91.0V - 92.0V - 93.0V - 94.0V - 95.0V - 96.0V - 97.0V - 98.0V - 99.0V - 100.0V

Meeting 2 (16/1/26)

- Weekly meeting every Tuesday from 1:30 pm to 2:30 pm

General Discussions:

- Establishing a starting point
- Farzad shared a product design handbook
- Framework: Flask (backend), Figma (frontend prototype), CSV + MySQL (data)
- Sketch design of UI by next week by Farzad

Roles:

- Frontend – Bradley, Farzad
- Data/Database – Jayden, Ashley
- Backend – Rumi, Pure, Akkshay, Carlin (flexible including the names above)
- Documentation – Akkshay
- Testing – Carlin, Pure

Frontend team:

- Sketch a web layout

Backend Team:

- Decide features required for prototype

Data Team:

- Create schema for a basic profile

Meeting 3 (20/1/26)

Dashboard Page (user):

- Profile icon and group icon in the top right
- Option to add action (as a sidebar). Collapsible with today's plan, view progress and challenges.
- Progress bar at the top with a weekly goal progress and a certain aim
- Individually tracked stats below the progress bar, (each stat can be compared to stats from the previous week). Can be clicked to open a more detailed overlay window. (With comparison either with other modes of transport or in a negative way)
- Less abstraction on the calculation, more focus on the extra details to make a bigger impact.

Group Logic:

- One profile can be the owner of a group, have the privileges for the group
- Person can choose to be part of a group or not

Backend notes:

- Moderators need to handle evidence
- Evidence needs to be photo based
- Open source software (cubtrek) to aid calculations (ask in workshop)
- Mobile web development (ask in workshop)
- Picture upload system
- For the first stage use Strava or something similar as evidence with a manual moderator for verification
- To be able to invite non users as a challenge organizer
- We need to figure out the challenges and how they work (who the challenge organizers are and how groups can join and participate in challenges)
- We will create the challenges as centralized organizers and then the challenge organizers (group leaders) will choose which challenges the groups participate in.

Meeting 4 (24/1/26):

Question for workshop:

- Do we have to provide a group reflection for the presentation

Database Design/ Core features:

- Creating challenges – only group leader and moderators
- Leaderboard – Local for prototype, within groups, maybe upgrade during phase 2, refresh weekly challenges
- Verification of CO2 stats – ActionType -> ActionLog -> Evidence
- Evidence – Photographs? URLs?
- Point system – local system

Backend team can start designing, based on V1.1 UML

VM:

- Accessible for all devices on the same network
- No need to worry about Security (username, password)
- Use SQLite to link to virtual machine ([SQLite —> VM guide](#))
- How to link Database to code?
- Create a VM through MySQL/ Google Cloud? (3 months free only)

Backend:

- Flask
- VScode

Tasks:

- Carlin, Pure and Jayden work together to create a VM at point this week
- Bradley and Ashely work on dashboard
- Pure work on cookies
- Farzad sort out web design
- Akkshay, Carlin and Rumi help with backend

Meeting 5 (27/1/26)

- Discussed progress and tasks that have been completed since the last meeting.
- Creating a basic formula for the carbon footprint that can be implemented immediately.
- The formulas will be implemented in the backend and not in the database, the final values will be stored in the database.
- User can register if they have a car, which can be used in the comparison system.
- Have an abstract point system.
- Have a more gamified interface
- Logging default actions
- Maybe have different points awarded based on the action difficulty

Feedback on Initial Design:

- Could have a leaderboard for each challenge
- Easy transparency for users to access information about the calculations for CO2 data
- Weekly streak bar to keep users engaged and motivated
- Have seasons so that users who join later are not disadvantaged
- Allow users to download their data for a season

Targets for Week 5 presentation:

- Account system
- Working front end
- 5 automated tests, one manual
- Location confined to Exeter
- Backend needs to be simplified slightly
- We need to show accurate data calculations

Tasks:

- Rumi: CO2e research
- Akkshay: Designing formulas for CO2e
- Carlin, Pure, Jayden: update requirements to push to github, update schema, start coding database

Meeting 6 (31/1/26)

- Rumi researched on CO2 emissions for transport
- Keep scope within Exeter
- Update varchar(50) to varchar(250)
- Action log (make it transport type dropping down to distance only, keep it simple for prototype)
- Walking, cycling, train, car. (diesel, petrol, hybrid, electric) – implement bus and diesel car only for prototype
- Travel, food, waste, energy (4 functions)
- API? To obtain distance between 2 stations

Frontend Discussion:

- Farzad – dashboard layout and design/formatting, Add challenges to dashboard design, show points awarded for each activity, demonstrate one evidence required, submission being approved/rejected with reasoning, action logging (design an extra card for action logging with stats and percentage progress contribution to the weekly activity)

Travel types:

Train:

- o Distance in km

Cycling:

- o Distance

Car:

- o Diesel
- o Distance

- Bradley - Login and register page code
- Ashley – code the 2 remaining cards for evidence upload stats

Meeting 7 (3/2/26):

Discussing presentation dates and details to include for the presentation

Updated figma design:

- User is now able to add actions (type of travel, etc.)
- User can upload evidence
- Pop up banner when the user submits the action
- The challenge page will show the current active challenges, there are tabs that can take the user into more detailed information/graphs about the challenges.
- Users can see a moderating list to see if their evidence has been approved or not and a list for groups they are part of

Defining the rules for how the user will add the actions to the challenges and how they apply:

- Add parameters in challenges so the actions are automatically added to challenges
- If the user does not need evidence, then its only counted towards their individual score (unofficial score)
- The action is automatically counted towards the challenge with evidence upload

Create a specified hierarchy for group leaders, moderators / challenges and their types

Add a filter for challenges page

There might be a need to add a category type into the table

Meeting 8 (10/2/26):

Front end:

- Future requirements table (must haves, should haves)
- Update the scrumboard for frontend

Database:

- Update the tables to autoincrement in workbench (Jayden)

Tasks:

- Finalizing one general report for CW1 by the end of this week (Akkshay)
- Ethics and licensing (Rumi)
- Section; meeting minutes, risk register (Farzad)
- Testing (Carlin + Pure)
- Slides (Carlin + Rumi)
- Cards by Thursday (Ashley and Bradley)
- Moderators: Users will have an extra page where they can approve or reject, providing reason for rejection from the lists of evidence that has been submitted.
- Backend and Frontend must be done by Sunday (Monday is presentation)

Meeting 9 (12/2/26)

Discussing current progress:

- User successfully registering
- Rumi will do some testing when the prototype is ready
- Successful logging of actions
- Small class to convert SQL commands to SQLite for automated tests
- Decide the future direction of the product after user feedback and tests
- No need to have fully-fledged front-end design for now, only the functionality
- Frontend should be ready to be linked to backend by Saturday
- Ensure distance tracking is properly implemented
- Moderator is a user with a higher privilege
- No rankings right now
- We don't have to be accurate with the numbers for CO2 calculations for the prototype
- For now, we will only use driving a car as the comparison point, for eg: walking compared with taking a car, train vs taking a car, etc.
- Each card will show weekly, monthly and total savings
- History table for moderations, showing approved and disapproved evidences
- No need to consider passenger per kilometer for the prototype
- Must disclose the type of emission calculations
- Public methodology page for calculations
- GDPR compliance
- Justification for risk assessment

SQLite Progress:

- SQLite database right now is for testing
- Users can register and use the login page

Meeting 10 (14/2/26)

- Backend and frontend will be linked soon
- Testing has been updated
- 2 more test files need to be completed
- Merge all the html and css components together
- Extra page for moderator where there is an icon on the top right which can be clicked to see new actions and evidence
- Presentation should be done by tonight
- Frontend recap meeting on Sunday

Meeting 11 (15/2/26)

- Add the confirmation prompt for CO2 factor

Testing functionality for the presentation:

- On the action input screen, user can input travel category, distance, etc.
- Dashboard update can be seen
- Challenges can be submitted

Presentation:

- Rumi will do the intro
- Bradely will present logging action and calculation plan
- Farzad will show the frontend design and dashboard update
- Carlin will show challenge submission
- Pure will show moderator review
- Database will be presented by Jayden
- Anti gaming checks will be presented by Akkshay
- Ashely will present risks
- Rumi will conclude
- **Subject to change**

Meeting 12 (18/2/26)

Submission:

- submission.txt (Rumi)
- team_members (Rumi)
- cw1_prototype_report.pdf (Akkshay)
- scrumboard_export.pdf (Akkshay)
- meeting_minutes.pdf (Akkshay)
- risk_register.pdf (Rumi)
- ethical_and_legal_considerations.pdf (Rumi)
- license_decision.pdf (Carlin)
- software_data_inventory.xlsx (Carlin)
- deployment_guide.pdf (Pure)
- testing_evidence.pdf (Pure)

Rumi will submit the coursework for the group

Meeting 13 (23/2/26)

Feedback:

- More interactive UI
- Transfer notion into Trello
- Add internal deadlines
- Make standardized README
- Work on separate github branches
- Include privacy and security in presentation
- Look more deeply into marking scheme and project brief
- Look into UAT (user acceptance testing)
- Connect questions to user requirements
- Seeded dataset

Meeting 14 (24/2/26)

Recapping yesterdays Surgery notes:

- README file has to be updated
- Sprint must be moved to a Trello Board
- Work on separate github branches
- Include privacy and security in next presentation

Back end will work on challenges functionality and logic

Frontend will be working on Design of action log, link the pages via buttons, create a navbar

Mobile compatible layout needs to be implemented

Convert the figma design into CSS

For energy action logging, the moderator could assign tasks daily or weekly which the users can chose to do if they want. This makes it easier as the moderator does the research on the task making it easier to implement.

Are we still going forward with the idea that the user can register a car

For food tracking the user will upload a picture of what they are eating or we can ask the user to upload their shopping list which we can analyze and then connect this with waste as well.

Meeting 15 (27/2/26)

- Energy research for electricity is complete
- Energy research scope is just electricity for now
- Discussed about the meter graph that tells the user how good their saving is
- Ashley, Bradley and Farzad will be designing the meter graph
- For food category we want to be transparent so include emission at each stage
- Rumi will be doing waste disposal research
- Current plan for waste disposal research is tell the user how much they are saving or not based on whether they recycle something and what type of waste they are throwing away.

Meeting 16 (3/3/26)

- User can only join 1 group
- Possible expansion to join multiple groups but focus on one for now
- User can be invited to groups by group admin
 - o Admin will search user user_id and invite them if ID exists
 - o User will receive a pop-up in challenges page about the invite
 - o Create a table /attribute in a table for pending invites
 - o Contains links to join groups/group_ids that have invited the user
- Assign markers account to have an invite to the group
- During presentation: show the process of inviting someone to a group
- Show off group functionality in the presentation as well
- Currently the owner cannot leave the group but maybe in the future make it so that the owner can transfer ownership and leave the group
- Discuss ranking within groups
 - o Special point allocation ?
 - o Average per user in group ?
- Discuss point allocation
 - o Allocate points per activity/saving ?
 - o Reward points based on streaks?
- Points are simply there for show in sprint 2, they wont be used yet but we can talk about it under future expansion how they could be used in a shop for special avatars, profile borders, etc.
- Notification page
 - o Group invites
 - o Challenge notifications
 - o Evidence Decisions
- Notifications can glow and once read wont glow
 - o We can discuss this more
- Change register page to include username
- Display username to other users
- Change login to require username

Meeting 17 (10/3/26)

Frontend:

- Designs for the action logs are about to be completed (3/4) done
- When you click on the 3 dots for action log, user can see calculations and formula
- We will ask the user what food related action they would like to record, give them some examples too. When they press submit they will see the option to upload the receipt as evidence, for now it will be manual by the moderator. If they don't upload a food receipt, we show a message saying we can quantify their savings
- Future implementation of food savings could be with Rydoo (receipt scanner)
- Users will be able to see the points that they have in reserve, which means this is the amount of points that will be added after their evidence is approved
- After the user enters the energy saving action, they will have the option to upload their evidence later or immediately. If they chose later, they will be reminded from the notification bar
- Add how we have reached the estimated saving number calculation in the travel log
- For waste, we can use the number of bin bags used and ask the user to input the size of the binbag (eg. 35L), for evidence they just upload the picture of the bag.
- Website should be deployed soon

Meeting 18 (16/3/26)

Deployment:

- Current issues being faced is already known by the IT department. If it is not fixed then we will present with an SSH

Presentation:

- In the spec
- First cover what has been done between sprint 1 and 2
- Handover includes what should be considered by the next group
- Very briefly, cover what has happened, no need to go too in depth

Meeting 19 (21/3/26)

Productivity meeting

Backend

- Implement group (presentation)
 - o Empty groups
 - o No owners ?
 - o One group per user
- Implement challenges (presentation)
- Implement energy (presentation)
- Implement food
- Implement waste

Frontend

- Card for waste (presentation)
- Card for food (presentation)
Add group addition for groups (presentation)
- Accept username (presentation)
- Add buttons to the moderator evidence (presentation)

Frontend/Backend

- Moderator makes challenges
- Implement group leaderboard
- Implement individual leaderboard
- Filter systems for leaderboards
 - o Date, range and category

Database/backend

- 500 dummy datasets
- At least 500 action logs across at least 60 users and 10 groups (synthetic data is acceptable)
- At least 8 challenges with at least 200 challenge submissions across them
- At least 80 evidence submissions seeded and at least 40 moderation decisions seeded
- A conversion factor for CO2e calculations (with sources included in your report)
- At 100 pre-seeded 'edge case' submissions to test anti-gaming checks and moderation

Meeting 20 (23/3/26)

Presentation:

Slide 1 – Title (Rumi)

Slide 2 – Problem & Solution (Rumi)

Slide 3 – Key Features (Rumi)

Slide 4 – System Architecture (Carlin)

Slide 5 – Emission Tracking (Carlin)

Slide 6 – Gamification & Challenges (Jayden)

Slide 7 – Testing (Jayden)

Slide 8 – Measurable Success Criteria (Akkshay)

Slide 9 – Ethics & Privacy (Akkshay)

Slide 10 – Risks & Mitigations (Rumi)

Slide 11 – Team & Development Process (Rumi)

Slide 12 – Deployment & Handover (Carlin)

Slide 13 – Future Work (Carlin)